CODESTREET 2026 · AMERICAN EXPRESS · ROUND 1

Autonomous Travel-Disruption Concierge

It re-books the flight, moves the hotel and the cab, and claims every duty of care DGCA already owes the traveller — in about eleven seconds, without a phone call.

~11 s
cancellation to a confirmed, paid-for trip

42 s
of the work already done before it happens

3
consent modes, one default-deny policy gate

100%
agent actions passed through OPA / Rego

PRESENTED BY

TEAM ZKD

- **M. Zayaan** — CE23B092
- **Krishna Satyam** — CE23B086
- **Dhawal Khatri** — CE23B063

Indian Institute of Technology Madras

Built on Indian regulatory and DGCA operational data. Every number carries its source.

One traveller. 180 minutes. Nobody closes the loop.

India flew 16.4 crore domestic passengers last financial year. Disruption is not an information problem — it is an execution problem.

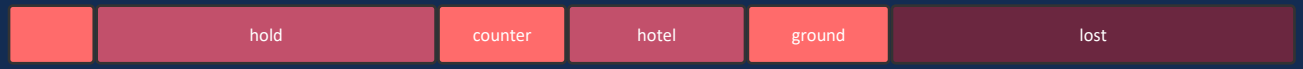
PRIYA · AMEX CARD MEMBER

MAA → DEL → LHR

Board meeting in London tomorrow · hotel booked · car booked

06:12 — Chennai leg cancelled. The DEL connection, the hotel and the car are all now wrong.

TODAY



180 min · 1 call-centre contact · duty-of-care never claimed

CONCIERGE



4 min · 0 calls · meals, hotel and transfer auto-claimed

48 of those 180 minutes are hold music — the most expensive minutes of the journey for the issuer.

17.2%

of flights arriving late

IndiGo OTP 82.8%, May 2026 — SpiceJet was 43% in March

₹4.24 cr

spent by Indian carriers on delay facilitation

in a single month — May 2026, DGCA report

2.4 lakh

passengers hit by delay or cancellation

1.77 lakh delayed + 63,723 cancelled, May 2026

₹22.2 cr

DGCA penalty on IndiGo

after 2,507 cancellations stranded 3 lakh flyers, Dec 2025

WHO IS ALREADY IN THIS MARKET — AND WHERE EACH ONE STOPS

Lumo

predicts delays, never transacts

Freebird

rebooks the air leg, nothing else

Kaiban

airline-side IROPS, optimises the carrier

Navan / TMCs

corporate perimeter, no benefits layer

AirSewa

government grievance portal — after the fact

THE SEAM

Prediction, rebooking, ground logistics and statutory duty-of-care are four separate industries. The traveller is the systems integrator between them — and only the issuer holds a relationship that spans all four.

SOURCES DGCA monthly operational reports, March & May 2026 (OTP, cancellation rate, passengers affected, facilitation spend); DGCA Annual Report FY2024-25 (16.4 cr domestic passengers); DGCA penalty order on IndiGo, Jan 2026; company product pages.

Five pipelines. One policy gate. Nothing acts alone.

Mapped directly onto the five tasks in the problem statement — each pipeline ships independently, neither horizontal layer is optional

01

Core Workflow

Detects the disruption and re-books flight, hotel and car as ONE distributed transaction.

02

Predictive Watcher

Forecasts the delay and the missed connection before the carrier announces it.

03

Cascading Logistics

Recomputes every downstream dependency — transfer, check-in, onward leg.

04

Conversational Fallback

Knows the moment it must stop, and escalates with full context attached.

05

Duty-of-Care Enforcer

Executes what DGCA already obliges the airline to give: meals, hotel, transfer, re-accommodation.

POLICY DECISION POINT — Open Policy Agent / Rego · default deny · this is what "within policy limits" actually means in code

DURABLE EXECUTION — Temporal.io · at-least-once activities, idempotency keys, compensating actions in reverse order · nothing is fire-and-forget

HOW THIS MAPS ONTO THE FIVE STATED TASKS

Task 1	P1 + P2	monitor live flight data, detect disruptions
Task 2	P1 + policy gate	autonomous rebooking within policy limits
Task 3	P4 + app	member-facing alerts, options, confirmations
Task 4	P1 + P3	airline, hotel and notification APIs
Task 5	metrics, slide 8	detection speed, rebooking success, notification clarity

Closed-loop network
issuer, network and acquirer at once — the agent can verify what it booked actually settled

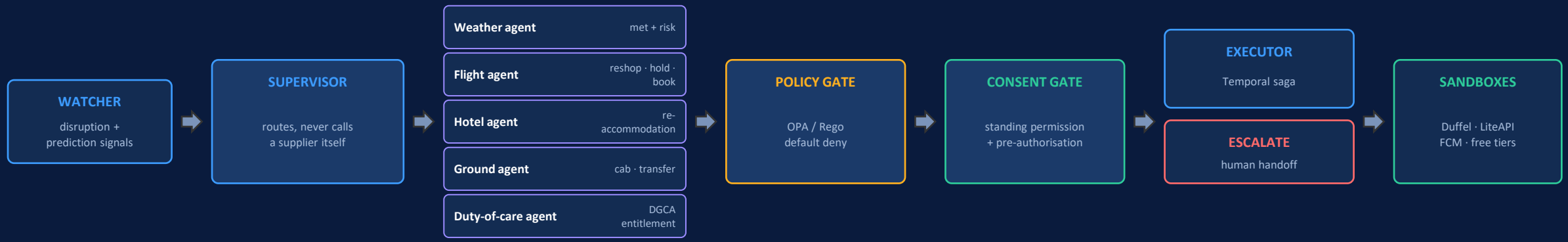
Entitlement is already a rule set
DGCA CAR Section 3 in India, card benefit terms elsewhere — a specification, not marketing copy

The agentic rails already exist
Amex ACE issues scoped tokenised credentials; vPayment locks a single-use card to an amount and a date

SOURCES Task list quoted from the Codestreet 2026 problem statement 'Autonomous Travel-Disruption Concierge'. Sabre agentic APIs / MCP server, 23 Sep 2025; Lumo API suite; Amex ACE Developer Kit, 14 Apr 2026; Amex vPayment; OPA (CNCF Graduated).

A supervisor graph, and consent collected before the panic

Sub-agents propose. Only the executor acts — and only after the policy gate and the consent gate both pass.



`TripState { disruption · pnr · consent · pre_auth? · candidates[] · policy_decisions[] · rejected_offer_ids[] · holds[] · confirmed[] · escalation? }`

1 · AUTOPILOT

Fix it and tell me after. Captured once, when the card is activated — not a toggle buried in settings.

Quiet window: notify → 90 s → proceed if silent.

2 · ASK ME FIRST

Plans, holds and presents — then waits. But consent gates SPENDING, not care: if the recovery costs nothing, silence books it rather than strand you.

Costs money and no answer? We stop and hold.

3 · PRE-AUTHORISED

Above 80% risk we ask in advance, while there are hours to think. Answer once and the cancellation needs no 90-second window at all.

Conditional: it fires only if the flight actually cancels.

EVERYTHING IS HELD BEFORE IT IS BOUGHT

Every leg is taken as a tentative hold with a release deadline. The agent confirms only after both gates pass; an unconfirmed hold expires for free. That one decision is what makes the 90-second quiet window safe, the rollback cheap, and tier B possible at all — a plan can be shown to the Card Member before a rupee moves.

HOW THE AGENTS ARE HARNESSED

- Sub-agents are tool-callers with no authority — each returns a schema-validated proposal.
- Only the executor node touches a supplier, through the Temporal saga.
- Consent tier is an OPA input, not a UI setting — one policy gates fare, cabin and who may confirm.
- Max 3 supervisor iterations, no cycle without progress, unroutable state exits to escalation.

SOURCES Supervisor / sub-agent pattern implemented on LangGraph. Consent tiers, the tentative-hold model and the harness rules are original to this submission. Sandbox availability verified: Duffel test mode and the LiteAPI sandbox key are free and self-serve.

Six layers — and one edge that deliberately does not exist

The agent proposes. It has no network route to an integration except through the control plane.

1 EXPERIENCE	React Native push + dashboard	Chat surface (fallback)	Wallet / pass update	Ops console audit view	
2 EDGE	API gateway + mTLS	AuthN / AuthZ OAuth 2.0	Per-credential rate limiter	Idempotency filter	
3 AGENT LAYER proposes only	Planner LangChain	Ranker + explainer	MCP tool clients	Grounding offer-ID	Escalation classifier
4 CONTROL PLANE authority lives here	OPA / Rego PDP default deny · decision log	Temporal.io durable workflows · saga			
5 INTEGRATION	Sabre agentic APIs / MCP	Lumo predictive	Hotel CSL + ground	Amex ACE + vPayment	FCM v1 data messages
6 DATA & AUDIT	Event bus	Feature store	Decision ledger	BigQuery KPI	

THE RECOVERY, IN SIX STEPS

01 **Disruption signal**
webhook / prediction

02 **Context assembly**
PNR · DAG · benefits

03 **Option generation**
reshop + lodging + ground

04 **Policy evaluation**
OPA admits or denies

05 **Saga execution**
compensation registered first

06 **Settlement + notify**
vPayment VAN, then FCM

deny at 04 → escalate to pipeline 04

invariant: every executed side effect ⇒ exactly one OPA allow decision ∧ exactly one registered compensation

Testable, not aspirational: a property test asserts the decision ledger and the compensation registry each hold one matching entry for every side effect in the workflow history.

SOURCES Original architecture. Component capabilities: Sabre developer hub; Lumo API reference; Amex ACE / vPayment; Firebase Cloud Messaging; Temporal.io; Open Policy Agent.

How an AI gets spend authority without becoming a liability

The model never touches a payment API. It emits a proposal; three independent layers decide whether it survives.

WHEN A BOOKING FAILS HALFWAY

A1	reserveVAN	ok
A2	bookFlight	ok
A3	bookHotel	ok
A4	bookGround	FAILS

compensations run in reverse (LIFO)

C4	cancelGround	no-op
C3	cancelHotel	confirmed
C2	voidFlight	in window
C1	releaseVAN	reversed

Traveller ends on the original itinerary. No orphaned hotel night, no held authorisation.

INDIAN LAW, COMPILED TO REGO

```
package concierge.dutyofcare
# DGCA CAR Sec.3, Series M, Part IV

owed contains "meals" if {
  input.delay_hours >= 2
}

owed contains "hotel_and_transfer" if {
  input.delay_hours >= 6
  input.overnight_window
}

owed contains "alt_flight_or_refund" if {
  input.delay_hours >= 6
}

# force majeure removes CASH,
# never the duty of care
cash_due if not input.force_majeure
```

Cancellation slab, same bundle: ₹5,000 / ₹7,500 / ₹10,000 by block time, or the booked fare — whichever is less.

FOUR INDEPENDENT LAYERS

L1 Grounding
no supplier offer ID, no booking

L2 Policy
OPA default-deny on every proposed action

L3 Credential
vPayment VAN locked to an amount and a date

L4 Network
closed-loop settlement + Agent Purchase Protection

An attacker must defeat all four. Two of them belong to the network, not to us.

Entitlement is data, not code. A second bundle encodes Amex card terms; Amex operates in 130+ countries, so the market changes the bundle — never the engine.

SOURCES DGCA CAR Section 3, Series M, Part IV, 'Facilities to be provided to passengers by airlines due to denied boarding, cancellation of flights and delays in flights' — as summarised by the Ministry of Civil Aviation, PIB release 1984936, 11 Dec 2023. Cancellation slabs corroborated across independent legal summaries. Rules are illustrative and must be reconciled with the current CAR text before production.

The model buys time. It does not decide anything.

A cancellation is a fact the airline files — we never need to guess it to act on it. The forecast exists to move work before the event.

If we cry wolf and the flight operates → it costs one API call, and the member never knows.

If we miss one and it cancels → recovery takes 53 s instead of 11 s. Still recovered. Neither case strands anybody, which is exactly why an imperfect model is safe to run in production.

FIVE WEIGHTED SIGNALS

34%	**Weather at both ends**	METAR/TAF against published runway minima — not a consumer forecast
26%	**Your aircraft's inbound**	A late inbound on a tight turn is how a cancellation is MADE
18%	**Airport congestion**	Movements in the ±30 min bank, plus any ATC flow restriction
14%	**This route's record**	DGCA per-carrier history, and our own observations
8%	**Time of day**	Delay accumulates through the operating day

AND WHAT WE DO AT EACH LEVEL

0–24%	**Watch only**	Re-score every 10 min. Your phone stays silent.	silent
25–54%	**Start preparing**	Assemble the trip, one search for the whole route, price the options. No hold, no spend.	silent
55–79%	**Prepare harder**	Evaluate the hold gate; pre-compute the policy verdict on every candidate.	silent
80%+	**Come and ask you**	Show the whole plan and the exact cost — while there are still hours to think.	WE NOTIFY
Cancelled	**Act**	Pre-authorised? Execute at once. Otherwise you get a real 90 seconds.	WE NOTIFY

Below 80% nothing pings you — a probability is not something you can act on, and an app that cries wolf gets muted. Production model is gradient-boosted trees on tabular features, calibrated, CPU-only.

SOURCES | Feature weights seeded from DGCA cause-of-cancellation splits; the prototype ships a transparent weighted model, NOT a trained one, and says so on screen. Bands, thresholds and the pre-authorisation rule are original to this submission.

The best time to ask is before anything has happened

WARM moves the searching off the critical path. Pre-authorisation moves the consent — the only part of a recovery that otherwise has to wait for a human.

WITHOUT PRE-AUTHORISATION

1. **Cancellation lands**
   the airline files it
2. **We notify, and wait**
   a real 90 seconds, under pressure
3. **Then we act**
   ~11 s of machine time

WITH IT

1. **You answered hours ago**
   calmly, with the full plan in front of you
2. **Cancellation lands**
   the airline files it
3. **We act immediately**
   no window, no human in the critical path

WHAT MAKES IT SAFE TO GIVE

Conditional

It fires only if the flight actually cancels. If it operates, the instruction expires unused and unnoticed.

Specific

It covers the exact plan you were shown. If any part is gone by then, the authorisation does NOT carry over — we come back and ask.

Bounded

Payment is a single-use card locked to that amount and that date. It cannot be reused or overspent.

Revocable

A flight you rejected can never be re-proposed — and that exclusion is a policy input, not a prompt instruction.

CONSENT GATES SPENDING — NOT CARE

If a member chose 'ask me first' and then does not answer, we book anyway WHEN THE RECOVERY COSTS THEM NOTHING. Permission exists to protect their money; leaving somebody at an airport overnight because they missed a notification is the worse outcome. If it would cost them, silence means stop — and the seats stay held.

SOURCES | Quiet-window mechanics map onto the RBI Additional Factor of Authentication e-mandate framework as a recognised pre-debit notification. The pre-authorisation model, the guard conditions and the spending-vs-care rule are original to this submission.

What we build on, and what we honestly cannot get

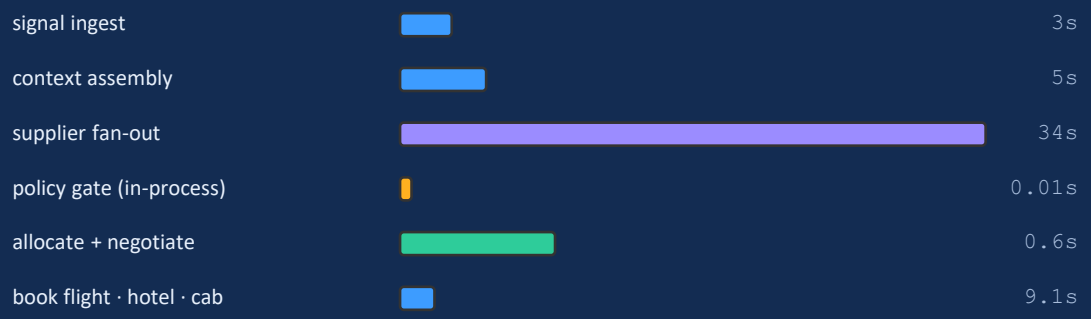
In this domain the API is the risk — so every dependency carries an access rating

STACK		ACCESS
Frontend	**React Native**	open
Backend	**Node.js / TypeScript**	open
Agent runtime	**LangChain**	open
Orchestration	**Temporal.io**	OSS + free tier
Policy engine	**Open Policy Agent / Rego**	CNCF, Apache 2.0
Air & hotel	**Duffel + LiteAPI sandboxes**	FREE · key in ~1 min
Amadeus	**Self-Service APIs**	DEAD — 17 Jul 2026
Agentic GDS	**Sabre Dev Studio test key**	free, self-serve
Predictive	**Lumo APIs**	commercial · mocked
Payments	**Amex ACE + vPayment**	select devs · mocked
Messaging	**Firebase Cloud Messaging v1**	open

Amadeus is listed on the challenge page — its free portal was switched off on 17 Jul 2026. Duffel and LiteAPI sandboxes give a real book AND cancel round trip, free.

WHERE THE TIME GOES — AND WHEN

53 s cold · 42 s of it moved BEFORE the event · ~11 s live



The 34 s fan-out runs BEFORE the cancellation, off the critical path. Of the ~11 s that remains, 95% is waiting on supplier APIs — the thinking is 0.6 s. There is no GPU on the critical path.

OUR BURST TEST IS A REAL EVENT

3 lakh

passengers stranded in 72 hours when IndiGo cancelled 2,507 flights and delayed 1,852 more, 3–5 Dec 2025. We size the system for that day, not for an average Tuesday.

Rate limits, not servers, are the ceiling.

WHAT WE CANNOT GET, AND WHAT WE DO INSTEAD

Amadeus self-service is switched off

→ build on Sabre, abstract the supplier

ACE is limited to select developers

→ mock behind a contract test

Lumo accuracy is a vendor claim

→ advisory only until back-tested

SOURCES PhocusWire 9 Feb 2026 + Amadeus developer notice (self-service portal decommissioned 17 Jul 2026); DGCA penalty order on IndiGo, Jan 2026, covering 3–5 Dec 2025; Sabre 23 Sep 2025; Amex ACE 14 Apr 2026; thinklumo.com. Latency figures are our engineering budget, not measurements.

Three value pools — and the arithmetic behind them

Normalised to 100,000 protected trips a year on Indian routes. Amber inputs are assumptions, and we say so.

POOL A

Contact deflection **$42.1K**

per 100K trips / year

Gartner global benchmark — an upper bound for India

POOL B

Billed business on Amex rails **₹8.7 cr**

per 100K trips / year

Recovery spend that today leaks off-network

POOL C

Retention & benefit utilisation **Δr × N × C**

formula, not a guess

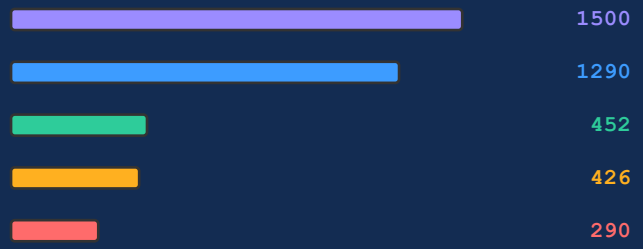
Third term is Amex-internal; we refuse to invent it

THE ARITHMETIC, IN FULL

100,000	17.2%	6,020	60%	$11.66	$42,116
protected trips	arrive late	call-centre contacts	deflected	saved each	POOL A
base	DGCA	35% assumed	target	Gartner	derived

RICE — BUILD ORDER

(Reach × Impact × Confidence) ÷ Effort



RICE ranks P2 first on reach, but we build to order and demo P1 + P4.

SUCCESS METRICS — THE THREE THE BRIEF ASKS FOR, DEFINED

Detection speed	`delay-ratio crossing → action`	**8 s, modelled**
Rebooking success rate	`same-day seat, constraint held`	**65.5% modelled**
Notification clarity	`FCM delivered ÷ accepted`	**≥ 97%**
Policy adherence	`actions with an OPA allow`	**100% hard gate**
Hallucinated bookings	`no valid supplier offer ID`	**0 by construction**

Outcome mix from a 250,000-case Monte Carlo at a fixed seed: A 52.6% · B 12.9% · C 27.6% · D 6.9%. Reproducible — python3 iropssim.py.

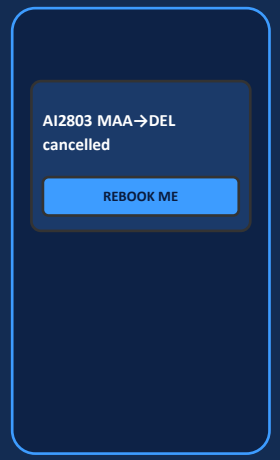
SOURCES DGCA monthly OTP (IndiGo 82.8%, May 2026) for the disruption rate; Gartner cost-per-contact benchmarks, flagged as a global upper bound pending Amex India telemetry; basket build-up ₹6,000 fare differential + ₹4,500 hotel night + ₹800 transfer, all assumptions; RICE per Intercom's published method.

Three surfaces — and the prototype is already built

Push notification → recovery dashboard → conversational fallback. If the traveller does nothing, the plan still executes.

1 PUSH

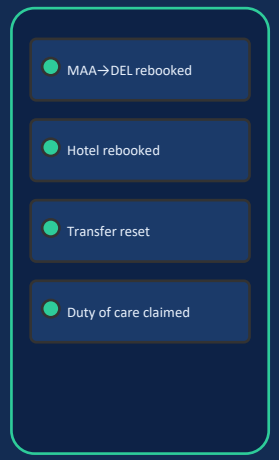
the only thing most users see



- FCM data message — the app renders the payload, not a generic banner
- One line of state, one line of action, one control

2 DASHBOARD

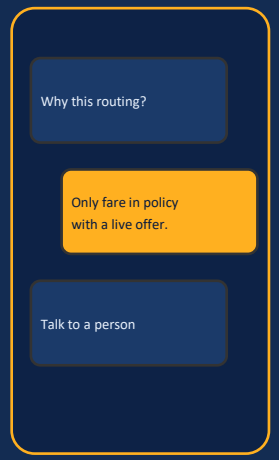
the proof surface



- Recovered trip timeline with per-leg status and cost
- Every decision carries a 'why' chip → its policy rule ID

3 CHAT

the bounded fallback



- Opens pre-loaded — the traveller never re-explains
- One tap to a human, who inherits the full handoff object

DELIVERY, ON THE PUBLISHED CALENDAR

PHASE 1 · VIRTUAL — **13–27 Jul**

NOW — This deck · architecture · Rego bundles · saga · AND a working prototype

Exit: submitted before 27 Jul, 23:59 IST.

FINALISTS ANNOUNCED — **7 Aug**

Top 6 teams named

MENTORING + PROTOTYPE — **7–21 Aug**

DONE EARLY — web app (8 routes) and an installable Android build, both driving the same engine. Links on the next slide.

Exit met: live rebooking, policy denials, LIFO rollback.

IN-PERSON FINALE — **25 Aug · Chennai**

Live demo of P1 + P4, with a recording as the fallback

Exit: prototype completion, business value, innovation, presentation.

BEYOND 25 AUG · Sabre onboarding · Lumo back-test · vPayment sandbox · one card product, India first

Kill criterion: if the Aug 7–21 build cannot produce a clean rollback on demand, the durability claim is unproven and we say so — rather than hide it.

SOURCES FCM data-message behaviour per Firebase message-type documentation (4,096-byte ceiling). Interaction design and delivery plan original to this submission; timeline per the published Codestreet 2026 schedule.

It is not a mock-up. It runs.

A web application and an installable Android build, both driving the same engine — generated inventory, a real policy gate, and a saga that rolls back.

WEB APP

8 routes

Flights · risk detail · the 80% early ask · live recovery · profile · permissions · history · how it works

ANDROID

installable APK

Release build, JS bundled — no dev server. MAX-importance notification channel with three action buttons

THE ENGINE

computed, not scripted

Inventory generated from route physics · default-deny policy · min-cost allocation · LIFO saga

THE MODEL

250,000 cases

Monte Carlo at a fixed seed behind every sim-tier number — python3 iropssim.py reproduces it exactly

SEE IT YOURSELF

Prototype · UI	the web application	[link to be added]
Prototype · APK	installable Android build	[link to be added]
Personas	four recovery scenarios	[link to be added]
System design	architecture, scalability, feasibility	[link to be added]
Success metrics	the 250,000-case model and its sources	[link to be added]

Known limits, stated up front: no trained model yet · no live supplier integration · payment mocked · API failure not modelled · the Android build is a subset of the web app.

SOURCES | Prototype and Android build original to this submission. Duty-of-care thresholds carry evidence tier 'deck' — the primary DGCA CAR text has not been re-retrieved for this build and must be reconciled before production.

THE ASK

Give the agent authority. Keep the authority governed.

Disruption has been treated as a notification problem for a decade — which is why every product in the category stops at telling you what went wrong. We treat it as an execution problem. Execution needs spend authority, spend authority needs consent, and consent needs governance. That is the whole submission.

Sabre developer onboarding	unblocks Pipeline 1 against live inventory
ACE Developer Kit access	replaces the payment mock with the real credential path
One card product, India first	a pilot large enough to re-base the model on Amex India telemetry

<15 min

from cancellation to a confirmed, paid-for, policy-compliant trip — with every duty of care already claimed.

Thank you.